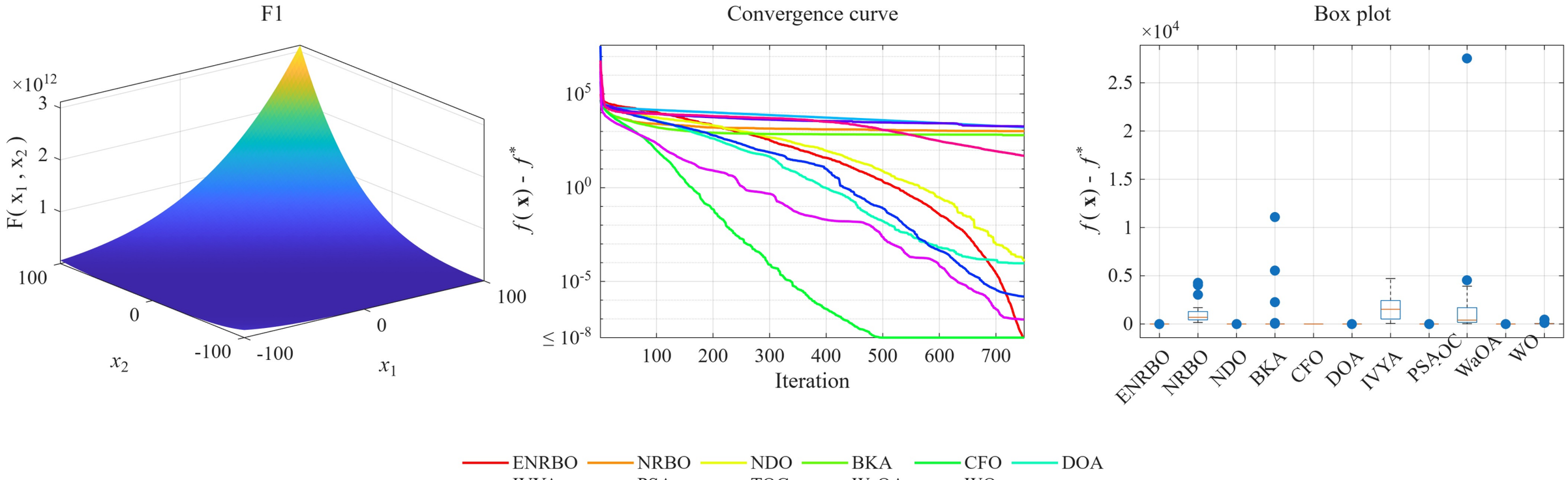

F1
×10^12
3
2
1
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^5
10^0
10^-5
≤ 10^-8
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
2.5
2
1.5
1
0.5
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO